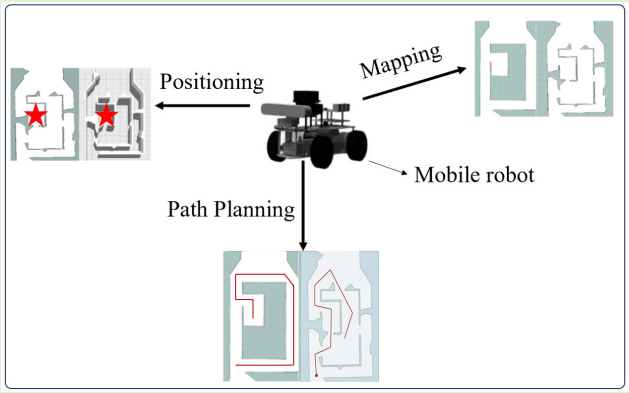
Autonomous Navigation for Mobile Robots in Emergency Rescue

Hongda Chen, Shanshan Sun, Zefeng Wang, Wei Shen, Jikai Yu, JeanMarie Nianga, Laurent Peyrodie, and Huajun Wang

***Abstract*—Emergency rescue robots are currently widely used in disaster relief scenarios such as earthquake response, mine collapses, and industrial explosions. Due to environmental challenges such as terrain variability and weak GPS signals, robots experience reduced localization accuracy and prolonged path planning, failing to meet the demands of emergency rescue operations. To address these issues, this study employs the Ackermann quick-response emergency mobile robot as the experimental platform. Initially, the application of the unscented Kalman filter (UKF) algorithm significantly enhanced the state estimation accuracy of the robot. Combined with the adaptive Monte Carlo localization (AMCL) algorithm, it achieved high-precision localization with an average accuracy error of less than ±0.036 m. Subsequently, the DAC-A* algorithm was proposed to address the issues with the traditional A* algorithm, such as excessive turning points and large turning angles, which delay the robot's arrival at the rescue site. Simulations conducted in MATLAB demonstrate that compared to the traditional A* algorithm, the DAC-A* algorithm reduces the time to reach the rescue destination by nearly 40%, substantially improving rescue efficiency. Finally, this research provides reliable technical support for disaster response, bearing significant practical significance and application value. Future enhancements through further algorithm optimization and exploration of more application scenarios could better meet the real needs of emergency rescue.**



***Index Terms*— A* algorithm, adaptive Monte Carlo localization (AMCL) algorithm, emergency rescue, path planning, robotics, and automation.**

I. Introduction

As URBANIZATION accelerates, the number of high-rise buildings has rapidly increased. However, in the event of earthquakes, the collapse of such structures often traps many individuals within unstable rubble, and aftershocks exacerbate the risk of secondary collapses, thereby compounding the challenges of rescue efforts and posing severe threats to life safety [1], [2]. In addition, earthquakes typically bring significant changes to terrain and indoor structures, such as ground fractures, interior walldisplacements, or collapses, and blocked passages. These alterations can render traditional rescue maps obsolete, particularly as discrepancies between original architectural blueprints and the actual damaged environments can lead to deviations in rescue route planning, significantly diminishing rescue efficiency [3], [4]. In the works of Wang et al. [5] and Zhou and Zhou [6], it is mentioned that although the inertial measurement unit (IMU) can provide immediate motion information, its cumulative error is relatively large, making it difficult to achieve high-precision positioning when used alone. Therefore, the IMU is typically required to be combined with other sensors, and the reliability and accuracy of data are improved through filter fusion technology to meet the rescue needs in complex post-disaster environments. Luo et al. [7] proposed an improved Unscented Kalman Filter (UKF) for IMU state estimation based on a modulation LSTM

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neural network, which can enhance the accuracy of IMU state estimation, but struggles to adapt to the rapidly changing, complex post-disaster environment. Xing et al. [8] utilized the extended Kalman filter (EKF) to integrate data information from IMUs, pressure sensors, and so on, to obtain high-precision positioning, but it is susceptible to marker failure caused by obstacle occlusion. Researchers like Donati et al. [9] and Li et al. [10] have employed multisensor fusion techniques to achieve centimeter-level localization accuracy; however, their reliance on GPS is unsuitable for signal-absent or weak indoor signal conditions. Visual sensors, though effective in normal conditions, are prone to malfunction in post-disaster settings due to dust, smoke, and obstructions, leading to ineffective image data and increased time for localization. Compared to traditional visual methods, the adaptive Monte Carlo localization (AMCL) algorithm integrates odometer data to optimize the Monte Carlo localization (MCL) process, improving the particle sampling and updating mechanism [11]. Nonetheless, the performance of AMCL heavily depends on the number and effectiveness of the particle updates, providing greater adaptability and efficiency in visually obstructed complex environments [12], [13]. In the context of path planning for mobile robots in post-disaster scenarios, while the traditional A* algorithm can provide globally optimal paths, its time efficiency and path smoothness are insufficient in settings with complex node traversals and frequent turning points [14], [15]. Researchers like Wang et al. [16] and Ou et al. [17] have improved the A* algorithm, reducing node traversals and road costs, yet turns remain frequent in complex environments. Huo et al. [18] proposed a novel method based on an improved ant colony algorithm and B-spline curves for smooth path planning of Ackermann mobile robots, which addresses the issues of slow convergence, low search efficiency, and poor path smoothness in the ant colony algorithm. However, the weight parameters for objective optimization are difficult to determine, imposing certain limitations in emergency rescue scenarios. Bascetta et al. [19] proposed a method enabling Ackermann robots to automatically manage parking and trajectory tracking issues, ensuring smooth changes in control signals, but ignoring the significant time consumption caused by right-angle turns in such scenarios for Ackermann robots. Nevertheless, when path direction changes are frequent, the optimal paths planned by the A* algorithm and ant colony algorithm are not always time-optimal, especially in time-critical emergency rescue scenarios where their shortcomings are particularly evident [20], [21].

To address the issues outlined above, this study employs the Ackermann quick-response emergency mobile robot as the experimental platform. The system architecture is depicted in Fig. 1. Initially, by integrating radar and odometer data, KartoSLAM efficiently generates accurate environmental maps. In addition, through loop closure detection mechanisms, the consistency and accuracy of the maps are further validated, enhancing the reliability of the map-building process. Subsequently, the UKF algorithm is used to fuse multisensor data to estimate the robot's pose. By integrating this with LiDAR observations, the estimated and observed poses are inputted into the AMCL algorithm to obtain a more precise global

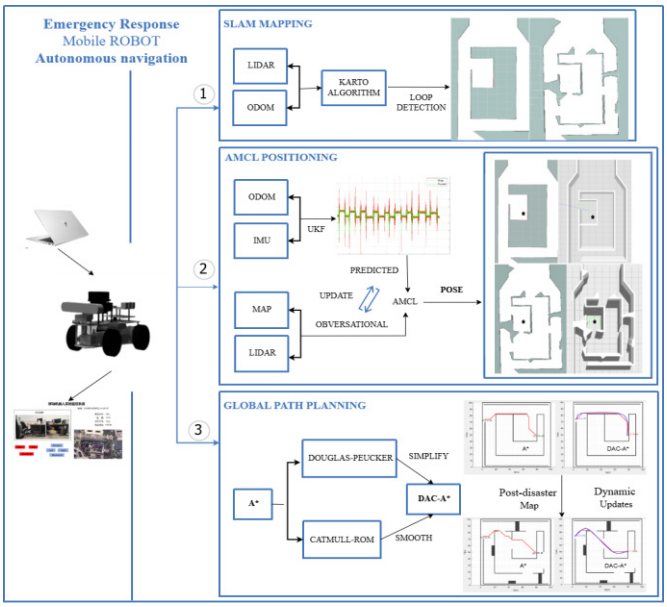


Fig. 1. System frame diagram.

pose in post-disaster scenarios. Finally, for path planning, this article improves the A* algorithm by incorporating the Douglas–Peucker path simplification and Catmull–Rom spline smoothing techniques, resulting in the DAC-A* algorithm. This algorithm reduces path node redundancy and enhances path smoothness, thereby achieving more efficient path planning and significantly reducing the time it takes for the robot to reach its target. This framework, which combines multisensor fusion for localization and path planning, provides reliable technical support for post-disaster emergency rescue scenarios.

The remainder of this article is organized as follows. Section II introduces the hardware architecture of the Ackermann mobile robot. Section III describes the principles of autonomous navigation for the mobile robot, including the global localization module improved through multisensor fusion with the UKF and the DAC-A* algorithm module for global path planning. Performance simulations and experimental results are analyzed in Section IV. The conclusion and future prospects are presented in Section V.

II. System Hardware Architecture

Building on the foundational aspects introduced earlier, this chapter details the overall system architecture of the study, which comprises hardware architecture, SLAM mapping, and autonomous navigation control. The hardware architecture encompasses the mobile robot's equipped sensors, controllers, drivers, and other hardware components, as well as their communication and integration methods. The autonomous navigation control is tasked with devising path planning and motion control strategies for the robot in complex environments, enabling real-time response and obstacle avoidance navigation in dynamic settings [22]. The hardware system of the mobile robot principally consists of sensors, control modules, drive modules, and communication modules. The structure of the hardware system is illustrated in Fig. 2. In addressing the challenges posed by increasing numbers of

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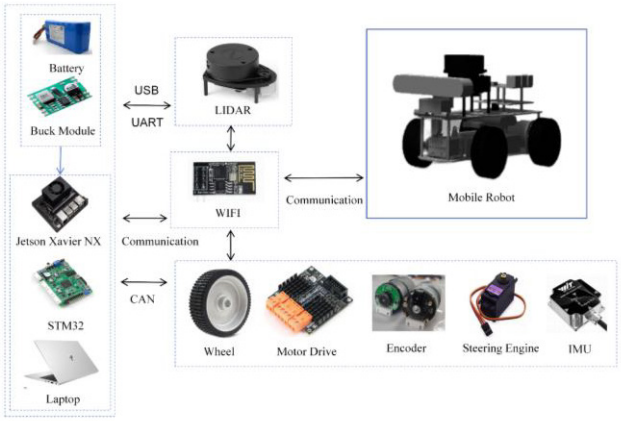


Fig. 2. Hardware system structure diagram.

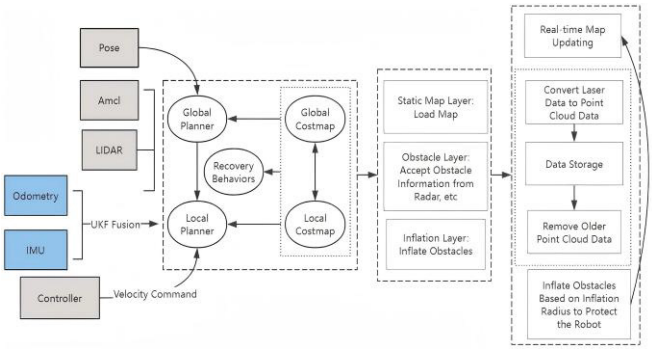


Fig. 3. Principle flowchart of autonomous positioning and navigation.

high-rise buildings in urbanized areas, the hardware system of the mobile robot designed in this study plays a pivotal role in disaster emergency response. The system consists of several highly integrated sensor modules, including encoders, radar, and IMU. These devices work collaboratively to provide the robot with real-time data about the damaged environment, supporting autonomous navigation and rescue operations within the debris. The STM32 microcontroller, serving as the sub-controller, directly connects to motor drivers and encoders, managing the robot's basic motion control to ensure stability and swift response in the dynamically changing disaster scenes. It is also responsible for collecting IMU data, which is crucial for maintaining balance and navigation in unstable rubble.

The Jetson Xavier NX, serving as the primary controller, processes complex data from radar and analyzes the layout of damaged structures and obstacles in real-time. Its high-performance computing capabilities enable the robot to understand and adapt to terrain and structural changes caused by earthquakes, which are vital for planning feasible rescue paths. Considering the practical needs of post-disaster rescue operations, the Jetson Xavier NX emits Wi-Fi signals through a wireless card, facilitating remote communication with the secondary units. This configuration allows operators to monitor and control the robot from a safe distance, especially when it enters buildings at risk of secondary collapse. Such remote operations reduce the risk to human operators while enhancing agile responses to disaster sites.

III. Principles of Autonomous Navigation of Mobile Robots

As a key technology in mobile robotics applications, autonomous positioning and navigation systems empower robots to reach predetermined locations autonomously while avoiding collisions. Depending on the operational environment, this technology can be categorized into three types: known environments, partially unknown environments, and completely unknown environments [23], [24]. This article focuses on how robots perform emergency rescue tasks in partially known indoor disaster building layouts when GPS signals are unavailable or unreliable. Often deployed in such settings following natural disasters like earthquakes and fires, rescue robots are tasked with identifying and navigating through partially damaged building structures. The detailed process of autonomous positioning and navigation is illustrated in Fig. 3, which outlines the comprehensive technological route from environmental perception and obstacle handling to path planning and execution, with a particular emphasis on ensuring the accuracy of robot positioning and navigation in complex post-disaster environments.

In the emergency rescue scenarios depicted in Fig. 3, vehicles equipped with Ackermann steering mechanisms often exhibit dynamic drifting during high-speed turning operations, which impacts the accuracy of odometer data. To address this issue, this study incorporates the UKF algorithm, which improves the accuracy of the AMCL by fusing odometer data with IMU outputs, replacing traditional odometer information. This precise localization technique ensures the efficiency and stability of path planning and execution. The navigation system includes global and local planners that generate global and local maps with cost grids, supporting complex path planning. The system also implements dynamic obstacle processing, where the static map layer is responsible for loading the base map, the obstacle layer processes obstacle information in real time, and the inflation layer expands the obstacle boundaries to ensure a safety distance. Specifically, the obstacle radius is inflated by 0.25 m through parameter configuration (i.e., inflation_radius $=$ 0.25 m), so that the grids around the obstacle are assigned increasing cost weights according to the distance. Finally, the inflated grid map is converted into a cost matrix, and the path planning algorithm selects the optimal path based on the cost. This advanced autonomous positioning and navigation system architecture not only enhances the operational stability of rescue robots in complex environments but also significantly improves rescue efficiency, enabling robots to quickly and safely locate and rescue trapped individuals in disaster sites like post-earthquake rubble. To assist readers who are not experts in this field in understanding the content of this article, Table I describes the acronyms of the algorithms used in the text as well as the functions of these algorithms.

A. Improved AMCL Positioning Principle

AMCL is an enhanced version of the MCL algorithm, incorporating a particle number self-adaptation mechanism to

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TABLE I
DESCRIPTION OF ALGORITHMS FUNCTIONS

Algorithm	Full Name	Function Description
AMCL	Adaptive Monte Carlo Localization	A probabilistic algorithm that estimates a robot's pose in a known map using weighted particles, adjusting particle distribution to adapt to dynamic environments and improve localization accuracy in noisy conditions [25,26].
A*	A Star	A heuristic path planner that finds the shortest path from start to target by balancing actual start cost (g-cost) and estimated target cost (h-heuristic), optimizing efficiency for real-time use [29].
DAC-A*	Douglas-Peucker and Catmull-Rom spline - A Star	An algorithm that reduces path node redundancy and enhances path smoothness by combining Douglas-Peucker and Catmull-Rom spline algorithms, enabling more efficient path planning and significantly shortening the time for robots to reach targets.

dynamically adjust the quantity of particles based on localization uncertainty, thus improving computational efficiency and localization accuracy [25], [26]. Although widely used in robotic localization, AMCL's performance is influenced by several factors, including the strong non-convex nature of laser rangefinder sensor models, odometer drift during turns, and the challenges of environmental complexity and unstructured features [27], [28]. To address these issues, this article employs the UKF to fuse odometer and IMU data, replacing traditional odometer inputs. The core algorithm involves the following steps.

The IMU provides angular velocity and acceleration data, the odometer offers incremental displacement information, and the UKF merges this data to achieve an accurate estimation of the robot's pose $\hat{X}_t$ The process begins with obtaining the current state of the robot $\hat{X}_t^-$ and the Kalman gain K_t

$$\hat{X}_t^- = f\left(\hat{x}_{t-1} u_t\right) = \begin{bmatrix} x_{t-1} + v_t \cos\left(\theta_{t-1}\right) \Delta t \\ y_{t-1} + v_t \sin\left(\theta_{t-1}\right) \Delta t \\ \theta_{t-1} + \omega_t \Delta t \end{bmatrix} \tag{1}$$

$$u_t = [v_t, \omega_t]^T \tag{2}$$

$$K_t = P_{xz} P_{zz}^{-1}. \tag{3}$$

Then, X_{t-1}, y_{t-1} represents the robot's position at time t − 1, Δt is the sampling time interval, v_t and ω_t are the linear and angular velocities at time t post-fusion, respectively. u_t denotes the control inputs for the robot's motion at time t, and θ_t is the robot's heading angle at the same time. P_{xz} is the covariance between state estimation and measurements, reflecting the correlation between estimation errors and measurement errors. P_{zz} is the covariance matrix of the measurements. From (1) to (3), it follows:

$$\hat{X}_t = \hat{X}_t^- + K_t \left(z_t - h\left(\hat{X}_t^-\right)\right) \tag{4}$$

where z_t is the liDAR observations at time t, $h(\hat{x}_t^-)$ is the liDAR observation function at time t.

Initialize N particles in the map and assign weights (5)

$$\omega_0^{[i]} = \frac{1}{N}. \tag{5}$$

Pose $\hat{X}_t$ (4) and speed u_t (2) after UKF fusion were used as inputs, the pose $x_t^{[i]}$ of particles is predicted and the weight $\omega_t^{[i]}$ of particles is updated

$$\omega_t^{[i]} \propto \omega_{t-1}^{[i]} \cdot p\left(z_t \mid x_t^{[i]}\right) \tag{6}$$

$$X_t^{[i]} = \begin{bmatrix} x_{t-1}^{[i]} + v_t \cos\left(\theta_{t-1}^{[i]}\right) \Delta t \\ y_{t-1}^{[i]} + v_t \sin\left(\theta_{t-1}^{[i]}\right) \Delta t \\ \theta_{t-1}^{[i]} + \omega_t \Delta t \end{bmatrix} + \eta_t \tag{7}$$

where $\eta_t \sim (0, Q_t)$ is a random variable that follows a normal distribution, the covariance matrix Q_t controls the noise size and direction, and $p(z_t|x_t^{[i]})$ is the likelihood function of the laser observation, evaluating how well the particle matches the real observation.

Finally, the particles are resampled according to the weight $\omega_t^{[i]}$ distribution to avoid particle degradation.

By integrating the UKF to fuse IMU and odometer data into pose estimation, this approach drives the particle filtering process with more accurate state and velocity information. Coupled with a resampling strategy, it effectively enhances the robot's localization accuracy and stability within the map.

B. DAC-A* Algorithm

The A* algorithm is a heuristic-based path planning method, considered an extension of Dijkstra's algorithm, widely applied in robotic path planning to enhance search efficiency and path optimization capabilities [29]. The evaluation function of the A* algorithm is represented as follows:

$$f(n) = g(n) + h(n) \tag{8}$$

where h(n) denotes the heuristic estimated cost from node n to the target node, typically calculated using two methods: Euclidean distance and Manhattan distance. Given the coordinates of the current node as (x, y) and the target node as (x_t, y_t), the Euclidean distance is defined by (9), and the Manhattan distance by the following equations:

$$h_E(n) = \sqrt{(x_0 - x_n)^2 + (y_0 - y_n)^2} \tag{9}$$

$$h_M(n) = |x_0 - x_n| - |y_0 - y_n|. \tag{10}$$

In this study, the Manhattan distance formula from the classical A* algorithm is adopted as the heuristic function. However, traditional A* often introduces numerous turning points when generating paths, not only increasing the path length but also making it unsuitable for Ackermann-steering robots due to complex turning requirements. The unique steering limitations of Ackermann robots prevent on-the-spot turns, leading in practice to routes and travel times far exceeding theoretical optima [30], [31]. To address this issue and enhance the efficiency and accuracy of path planning, this article introduces a new path optimization method named DAC-A*. The DAC-A* algorithm incorporates the Douglas–Peucker technique to reduce unnecessary turning points and employs the Catmull–Rom spline algorithm for path smoothing. This improvement not only optimizes computational efficiency but also ensures path smoothness, aligning more closely with the operational needs of Ackermann-steering robots.

The Douglas–Peucker algorithm reduces the complexity of a dataset by recursively reducing the number of path points

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while maintaining the overall shape [32], [33]. The steps are as follows: select the start point A (x_0, y_0) and the endpoint B(x, y) from the path, then find the point $P(x_p, y_p)$ that is furthest from the line segment AB. If the distance D from point P exceeds a predetermined tolerance, it is considered that these points are significant for the path shape, resulting in a simplified path. The formula for calculating D by the following equation:

$$D = \frac{|(y - y_0) \cdot x_p - (x - x_0) \cdot y_p + x \cdot y_0 - y \cdot x_0|}{\sqrt{(x - x_0)^2 + (y - y_0)^2}}. \quad (11)$$

Upon the basis of the simplified path, the Catmull–Rom spline, a method for generating smooth and continuous curves by interpolation, ensures that the curve passes through all key points [34]. The formula for the curve interpolation point Q(t) is given as follows:

$$Q(t) = 0.5 \begin{bmatrix} 2P_1 + (-P_0 + P_2)\, t \\ + (2P_0 - 5P_1 + 4P_2 - P_3)\, t^2 \\ + (-P_0 + 3P_1 - 3P_2 + P_3)\, t^3 \end{bmatrix}. \quad (12)$$

By incorporating the Douglas–Peucker path simplification technique and Catmull–Rom spline smoothing into the DAC-A* algorithm, this approach not only efficiently removes redundant path points and smooths the path curves but also significantly enhances the overall applicability and operational efficiency of the path planning process. This provides robust support for the precise and efficient operation of Ackermann-steering robots. Algorithm 1 illustrates the pseudocode of the proposed algorithm.

Algorithm 1 Douglas-Peucker Algorithm

```
function DouglasPeucker(points, tolcrance)
    dmax ← 0
    inder ← 0
    for i ← 2 to length(points) - 1 do
        d ← PerpendicularDistance(points[i], points [1], points[end])
        if d > d max then
            index ← i
            dmax ← d
        end if
    end for
    if dmax > tolerance then
        first half ← DotglasPEUCKER(points [1:index], tolerance)
        second half ← DouglasPeucker(points indexend, tolerance)
        simplified_path ← Combine(first_half, second_half) ≫ Merge results,
  avoiding duplicates
    else
        simplified_path ← [points ] ], points[end]] p Retain only the start
  and end points
    end if
    return simplified_path
end function
function PerpendecularDistance(point, lineStart, lineEnd)
    if lineStart = lincEnd then
        return EuclideanDistance(point, lineStart)
    end if
    line ⇇ lineEnd - lineStart
    t ⩽ DotProduct (point-lineStart,line) / DotProduct(line, IMe)
    t ← Clanp(t, 0, 1)   ▷ Ensure t is in the range [0, 1]
    projection ← lineStart +t× line
    return Euclidean Distance(point, projection)
end function
```

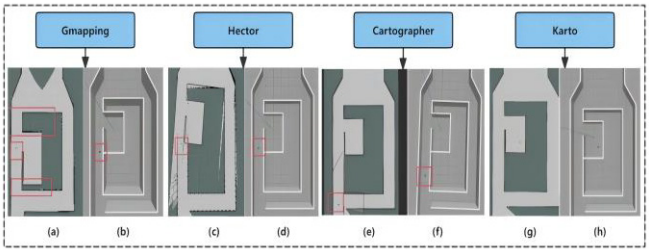


Fig. 4. Mapping performance comparison of four 2D-LiDAR-based SLAM algorithms. The figures contrast the generated maps indicating the estimated robot positions (left) against the actual robot positions in the environment (right). (a) and (b) Gmapping. (c) and (d) HectorSLAM. (e) and (f) Cartographer. (g) and (h) KartoSLAM.

IV. Result and Discussion

This study employs an Ackermann mobile robot with dimensions of 23 × 19 cm, operating in a corridor width of 150 cm, primarily to simulate real-world rescue scenarios and test the robot's adaptability and efficiency in constrained environments. The electronic hardware configuration includes a Windows 11 operating system and an Ubuntu 18.04 virtual machine, a 2.4 GHz Intel i7-13620HH processor, and 16 GB RAM, along with a Jetson Xavier NX running Ubuntu 18.04. The Jetson Xavier NX is particularly suited for high-performance computing embedded systems and artificial intelligence applications. It is equipped with up to 384 NVIDIA Volta architecture Tensor Core GPUs, a 6-core NVIDIA Carmel ARMv8.2 CPU, and high-speed 128-bit 8 GB LPDDR4x memory, capable of delivering up to 21 TOPS of processing power.

In the simulation environment, this study conducted map construction tests using a uniform speed (0.3 m/s) for common 2D-LiDAR-based SLAM mapping methods—Gmapping, KartoSLAM, HectorSLAM, and Cartographer [35], [36], [37]. Each algorithm underwent five independent tests to ensure the reliability of the results. To showcase the performance of each algorithm, representative test results were selected and detailed in Fig. 4, with the following analysis: The Gmapping algorithm exhibited a mapping error precision of approximately 0.7 m. By comparing the robot's position in Fig. 4(a) and (b), it is evident that in Fig. 4(b), the actual position remains at the intersection, whereas in Fig. 4(a), the map already shows the robot at the entrance of another room. HectorSLAM failed to complete a full map, as shown in Fig. 4(c), where the robot had not yet reached the intersection, while in Fig. 4(d), the actual position indicates that the robot had crossed the intersection. The Cartographer algorithm has a high speed requirement, accepting a maximum speed of 0.12 m/s, with Fig. 4(e) displaying the map generated at a speed of 0.09 m/s: in the figure, the robot has not yet passed through the horizontal corridor, whereas in Fig. 4(f), the actual position shows the robot has turned into the vertical corridor. In contrast, the KartoSLAM algorithm, even at a higher speed [0.5 m/s as shown in Fig. 4(g)], managed to perform loop closure detection with mapping error precision controlled at the millimeter level. Overall, the KartoSLAM algorithm was chosen to create a complete map and validate the enhanced precision of the improved AMCL algorithm.

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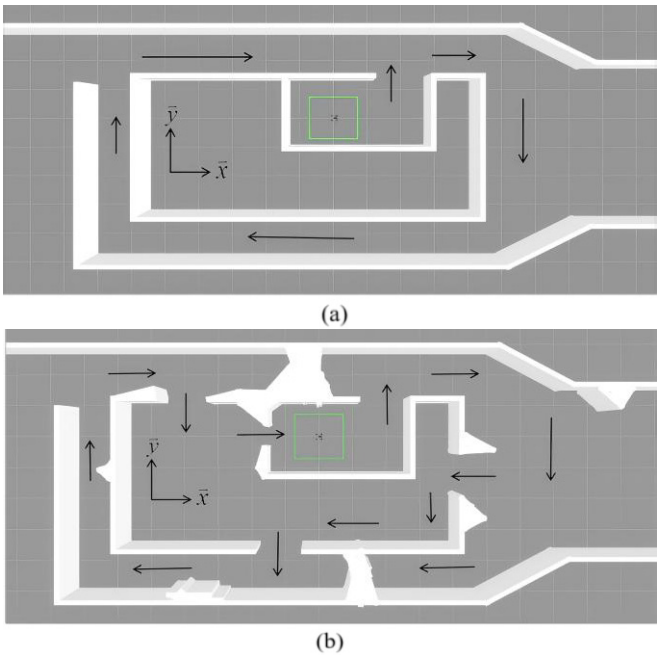


Fig. 5. Mobile robot initial position and test scene. (a) Not affected by disasters. (b) Post-disaster environment.

A. Improved AMCL Verification

Building on the previously established mapping algorithm, this section conducts an in-depth experimental evaluation and analysis of its accuracy and performance in mobile robot localization tasks, aiming to comprehensively assess its practical applicability and suitability. For this purpose, a 3-D map model was constructed in the Gazebo simulation environment, based on the seventh-floor circular corridor of the Student Innovation and Entrepreneurship Center at Huzhou University. An Ackermann mobile robot model equipped with encoders, IMU, and laser lidar was imported using the Insert module to simulate the robot's positioning performance in a real environment. The specific simulation environments (building office environment and post-disaster environment) are depicted in Fig. 5. In Fig. 5, each grid in Gazebo represents an actual size of 1 × 1 m; the squares indicate the robot's initial position, with the search path following the mapping direction shown in Fig. 5 (starting and ending at the same point). The primary goal of the experiment is to evaluate the localization error by comparing the differences in the longitudinal and lateral coordinates between the robot's current position (x_n, y_n) calculated by the AMCL algorithm and its actual position (x, y) in Gazebo, thereby verifying the algorithm's localization accuracy in specific environments.

The real-time status of the mobile robot is displayed through the ROS visualization tool Rviz, with the actual position acquired through the Gazebo simulation environment, recorded as (x, y); concurrently, the current estimated position displayed in Rviz is recorded as (x_n, y_n). During the experiment, this study utilized both the original AMCL algorithm and an improved version of the AMCL algorithm to perform coordinate measurements at 20 random test points on the initial map of the building office and its updated map after changes.

TABLE II
COMPARATIVE DATA TABLE OF THE TWO ALGORITHMS

Algorithm	Lateral Deviation(cm)		TMD (cm)	Vertical Deviation(cm)		LMD (cm)	Standard Deviation Ratio
	Maximum Deviation (cm)	Minimum Deviation (cm)		Maximum Deviation (cm)	Minimum Deviation (cm)		
AMCL	14.9	1.5	7.6	18.5	0.7	6.3	0.75
The Improved AMCL	8.4	0.2	3.6	8.3	0.1	2.7	0.97

The detailed data from these measurements are summarized in Table II, which is used to assess the global localization accuracy of both algorithms.

In this analysis, the transverse mean deviation (TMD) represents the average horizontal deviation distance after removing the highest and lowest deviation distances from 10 tests. The longitudinal mean deviation (LMD) represents the average vertical deviation distance after excluding the highest and lowest deviation distances from 10 tests. The standard deviation ratio is defined as the ratio of the standard transverse deviation to the standard longitudinal deviation, where,

Standard lateral deviation can be expressed as $((1/n)\sum_{i=1}^{n}(\text{Lateral deviation} - \text{TMD})^2)^{1/2}$, Standard Vertical deviation can be expressed as $((1/n)\sum_{i=1}^{n}(\text{Vertical deviation} - \text{LMD})^2)^{1/2}$.

Table II illustrates the following: The original AMCL algorithm had a maximum horizontal deviation of 14.9 cm and a minimum of 1.5 cm, while the improved AMCL algorithm had a maximum of 8.4 cm and a minimum of 0.2 cm. In terms of longitudinal deviation, the original AMCL algorithm showed a maximum of 18.5 cm and a minimum of 0.7 cm, whereas the improved version recorded a maximum of 8.3 cm and a minimum of 0.1 cm. For the original AMCL algorithm, the TMD was 7.6 cm and the LMD was 6.3 cm, both significantly reduced to 3.6 and 2.7 cm, respectively, in the improved algorithm. Overall, the standard deviation ratio for the original AMCL was 0.75, and for the improved version, it was 0.97. A standard deviation ratio closer to 1 indicates a more similar distribution of horizontal and vertical deviations, suggesting that the magnitude of error fluctuations across these two dimensions is similar. This reflects a uniformity in the deviations in both directions, which helps enhance the vehicle's stability and accuracy, ensuring that task execution in post-disaster environments is not unduly disrupted by unforeseen circumstances.

Subsequently, we simulated a more complex hospital environment, enhancing the challenge of localization by setting up intricate layouts within the hospital; for the post-disaster hospital scenario, we specifically designed collapsed structures and updated exit routes, among other map changes. The robot needs to be able to update its location information in real-time and effectively adjust its route to avoid obstacles, further validating the global localization precision of both algorithms in a more complex hospital setting. The specific simulation environment (hospital environment and post-disaster updated environment) is depicted in Fig. 6, with the pentagon marking

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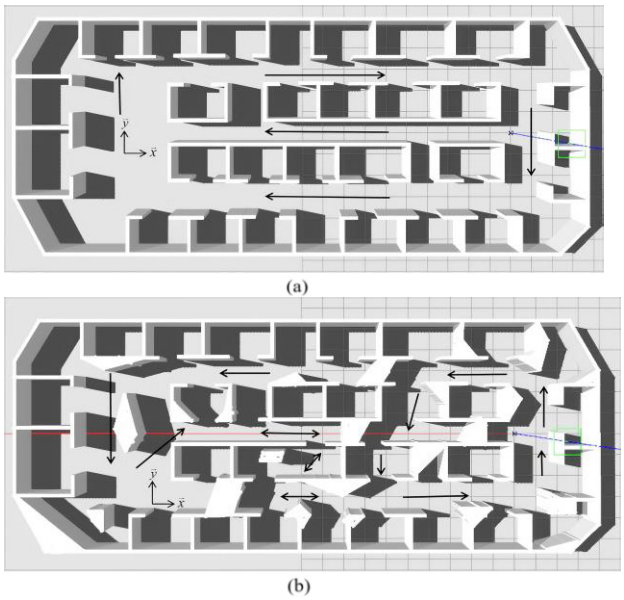


Fig. 6. Mobile robot initial position and more complex test scene. (a) Large supermarket. (b) After the destruction of the large supermarket.

TABLE III
COMPARATIVE DATA TABLE OF THE TWO ALGORITHMS

Algorithm	Lateral Deviation(cm)		TMD (cm)	Vertical Deviation(cm)		LMD (cm)	Standard Deviation Ratio
	Maximum Deviation (cm)	Minimum Deviation (cm)		Maximum Deviation (cm)	Minimum Deviation (cm)		
AMCL	16.3	1.4	5.7	22.9	3.3	6.6	0.76
The Improved AMCL	7.1	0.1	2.7	6.9	0.2	2.7	0.95

the robot's initial position, and the search path following the mapping direction shown in Fig. 6 (starting and ending at the same point). The main objective of the experiment is to evaluate the localization error by comparing the differences in horizontal and vertical coordinates between the robot's current position (x_i, y_i) computed by the AMCL algorithm and its actual position (x, y) in Gazebo, thereby verifying the algorithm's global localization accuracy in complex environments.

The mobile robot's actual position is recorded in the Gazebo simulation environment as (x, y), with the current estimated position recorded in Rviz as (x_i, y_i). During the experimental process, this study employed both the original AMCL algorithm and the improved AMCL algorithm to perform coordinate measurements at 10 random test points on the initial map of a large shopping mall and its updated map. The detailed data from these measurements have been compiled in Table III.

According to the data in Table III, the original AMCL algorithm had a maximum horizontal deviation of 16.3 cm and a minimum of 1.4 cm, while the improved AMCL algorithm reduced these figures to a maximum of 7.1 cm and a minimum of 0.1 cm. In terms of longitudinal deviation, the maximum and minimum values for the original AMCL were 22.9 and 3.3 cm, respectively, whereas the improved version reduced the maximum deviation to 6.9 cm and the minimum to 0.2 cm. In addition, in the metrics of TMD and LMD, the original AMCL algorithm recorded 5.7 and 6.6 cm, respectively, while the improved AMCL algorithm significantly lowered these to 2.7 cm each. Overall, the standard deviation ratio for the original AMCL was 0.76, which increased to 0.95 in the improved version, with a ratio closer to 1 indicating more similar fluctuations in horizontal and vertical deviations. These changes demonstrate that the improved AMCL algorithm not only significantly reduces the magnitude and variability of deviations but also aligns the distribution of errors in horizontal and vertical directions, thereby enhancing the mobility stability and operational precision of mobile robots in post-disaster environments, ensuring high task execution stability and reliability even under sudden and uncertain conditions.

B. DAC-A* Algorithm Verification

Next, to validate the effectiveness of the improved DAC-A* algorithm, we conducted simulations on the MATLAB R2022b platform, obtaining a more rapid and efficient optimal path planning solution. Within an 80 × 80 m environmental grid map, black represents impassable walls, while white grids denote navigable areas, with each grid measuring 1 × 1 m. Fig. 7 presents a comparative simulation of the routes before and after disasters using the A* algorithm and the DAC-A* algorithm.

From Fig. 7, it is evident that the paths in Fig. 7(a) and (b) are generated on post-disaster maps using the A* and DAC-A* algorithms, respectively, with the path in Fig. 7(b) appearing smoother near the endpoint. This indicates that the improved algorithm optimizes the path near the target point, reducing sharp turns, which is advantageous for vehicles with Ackermann steering mechanisms, as it can decrease the number of turns and enhance driving efficiency. Paths in Fig. 7(c) and (d) are generated based on post-disaster maps, with the route in Fig. 7(c) managing to avoid obstacles but having sharper corners, which may require the Ackermann vehicle to decelerate or make larger turning angles. In contrast, the path from the improved algorithm in Fig. 7(d) is smoother at turns while avoiding obstacles, benefiting the Ackermann vehicle by maintaining higher speeds safely. Path Fig. 7(e), compared to Fig. 7(a) and (c), navigates through a larger and more complex environment with significant corner changes, while path Fig. 7(f) shows noticeably fewer turning points and smoother transitions compared to Fig. 7(e). In summary, the DAC-A* algorithm provides smoother path planning, especially in post-disaster maps, crucial for Ackermann vehicles that need to efficiently avoid obstacles and quickly reach their destination. In addition, smoother paths can reduce the frequency of steering, decrease mechanical wear, and improve emergency response times.

To visually verify the time efficiency of the improved algorithm, both the A* and DAC-A* algorithms were applied for path planning on the original and updated post-disaster maps. Simulations were conducted with the Ackermann vehicle traveling at a speed of 5 m/s along the entire route, but slowing to 0.5 m/s at each turn. Efficiency was evaluated by

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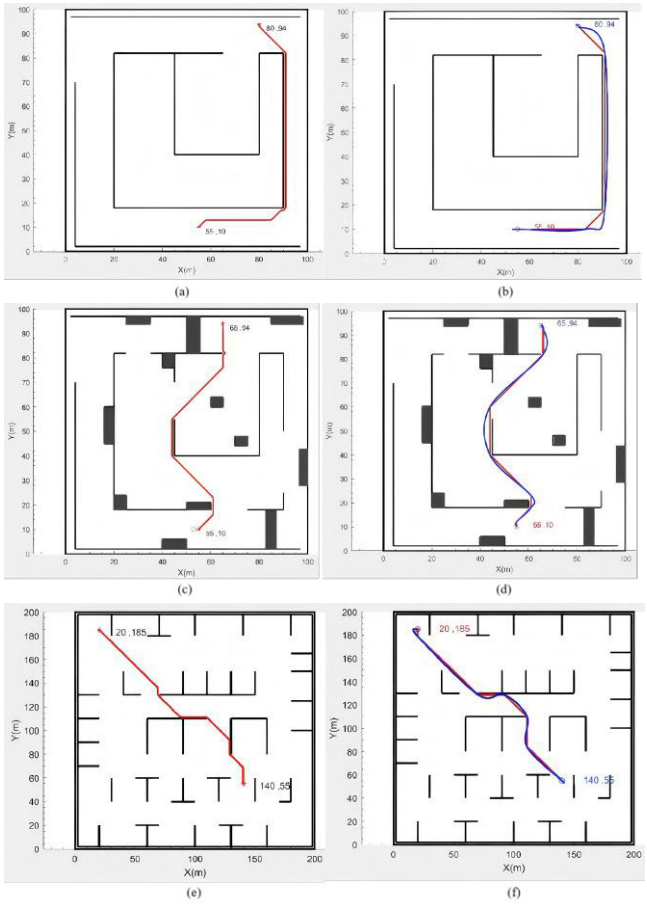


Fig. 7. A* and DAC-A* algorithm path effect. (a), (c), and (e) A* algorithm path effect in three different environments. (b), (d), and (f) DAC-A* algorithm path effect in three different environments.

recording the total time and distance required to complete the route. Each algorithm was simulated three times in each map environment (with different start and end points) to ensure data reliability. The corresponding simulation results are displayed in the bar charts in Fig. 8. Integrating observations from Figs. 7 and 8, it is evident that as the path complexity increases, the time required also lengthens. Although the A* algorithm may cover shorter distances, its numerous turning points make it more suitable for simpler scenarios or problems with low complexity. In contrast, the DAC-A* algorithm, despite having longer path lengths than the A* algorithm, takes the shortest time, demonstrating a better balance between efficiency and effectiveness, making it more suitable for complex post-disaster rescue settings. Specifically, in Fig. 8(a) and (c), the A* algorithm averaged a total distance of 82.97 m and a total time of 31.91 s on the original map, while the DAC-A* algorithm averaged a total distance of 91.82 m and a total time of 18.36 s. Although the total distance increased by 10.67%, the total time was reduced by 42.46%. In Fig. 8(b) and (d), on the post-disaster updated map, the A* algorithm averaged a total distance of 96.7 m and a total time of 39.23 s, while the DAC-A* algorithm averaged a total distance of 102.61 m and a total time of 20.52 s. Here, the total distance increased by 6.11%, but the total time decreased by 47.69%.

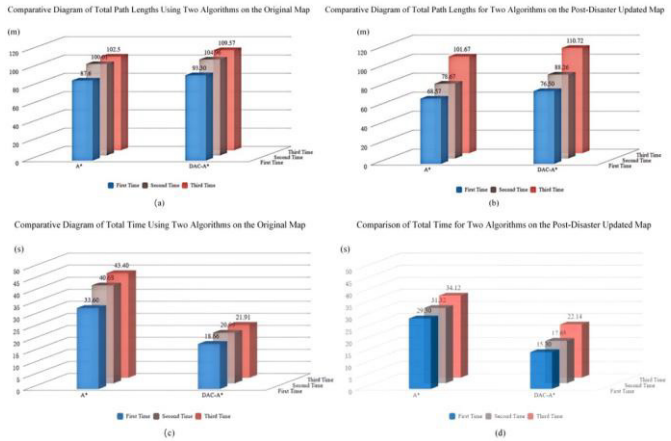


Fig. 8. Bar chart of total distance and time used by A* and DAC-A* algorithms. (a) A* (left) and DAC-A* (right) total distance in the original environment. (b) A* (left) and DAC-A* (right) updated total environmental distance. (c) A* (left) and DAC-A* (right) total time in the original environment. (d) A* (left) and DAC-A* (right) total environmental time after update.

Overall, despite the DAC-A* algorithm showing about a 10% increase in path length, it exhibits significant advantages in time efficiency, achieving at least a 40% reduction in time. This significant improvement means that while avoiding obstacles, the algorithm can compute feasible paths more quickly, greatly enhancing the efficiency of path planning. Particularly in emergency rescue scenarios like post-disaster environments where time is often critical to the success of the mission, the characteristics of the DAC-A* algorithm make it highly suitable for application in such complex, rapid-response settings. By reducing path calculation and execution times, this algorithm plays a crucial role in disaster response and crisis management.

V. Conclusion

To enhance the efficiency of emergency rescue operations, especially during the critical rescue window following disasters, this study developed a fast emergency rescue mobile robot based on the Ackermann steering mechanism. This robot utilizes KartoSLAM mapping technology with loop closure detection, significantly improving the accuracy and reliability of the maps produced. Moreover, by integrating the UKF algorithm to process odometer and IMU data, this research optimized the AMCL algorithm, achieving global positioning accuracy within ± 0.036 m, ensuring precise mapping and localization capabilities in complex environments. In addition, this thesis introduced the DAC-A* algorithm for path planning of Ackermann mobile robots, validated through multiple simulation experiments. The results demonstrate that compared to the traditional A* algorithm, the DAC-A* algorithm shows superior path smoothness and planning efficiency, reducing total time by 40%, and offering a novel solution for efficient path planning in practical applications. These findings provide theoretical support and practical references for autonomous navigation and localization technologies in post-disaster emergency rescue.

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The validation of DAC-A* was conducted in a MATLAB environment, where all conditions and parameters were precisely controlled, typically representing ideal scenarios without considering limitations such as the steering angles of Ackermann vehicles, path execution errors that may occur during actual movement, and terrain constraints where a robot may traverse rough terrain and experience positional deviations due to slippage. Environmental uncertainties, such as insufficient traction to climb over obstacles, inability to cope with smooth surfaces/water-logged areas, or impaired visibility, further necessitate the algorithm's capability to compute alternative routing strategies. These factors could lead to discrepancies between planned results and actual execution. Therefore, future research will further integrate real-world physical environments and introduce hardware experimental platforms to comprehensively validate the performance of the algorithm, ensuring its effectiveness and stability in real-world scenarios.

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